INTERNATIONAL JOURNAL OF HYDROGEN ENERGY 47 (2022) 39470–39485

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Review Article

Structural engineering of rare-earth-based perovskite electrocatalysts for advanced oxygen evolution reaction

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HIGHLIGHTS

- Recent advance in structural engineering of perovskites for OER is reviewed.
- The crystal structure and synthesis method of perovskites are discussed.
- Catalytic activity enhancement strategy of perovskites is reviewed systematically.
- A/B-site substitution, composite and morphology engineering of perovskite are involved.
- Challenges and opportunities confronting perovskites in superior OER are discussed.

ARTICLE INFO

Article history:
Received 27 May 2022
Received in revised form
11 July 2022
Accepted 13 September 2022
Available online 8 October 2022

Keywords:
Water splitting
Oxygen evolution reaction
Perovskite oxides

ABSTRACT

Oxygen evolution reaction (OER) acts an important role in electrochemical water splitting which converts sustainable and renewable energy into hydrogen fuel and the proper electrocatalysts are significant to the high-efficiency electrochemical water splitting. Rare-earth-based perovskites are promising catalysts for OER because of the low cost, the abundance of earth reserves, and coordinated electronic structures. However, their intrinsic OER property is still far from satisfactory from the viewpoint of industrial application, especially the electrocatalytic capability as well as stability under alkaline conditions. Herein, the recent advance in structural engineering of rare-earth-based perovskites for OER is reviewed comprehensively. The electrochemical characteristics enhancement strategy of rare-earth-based perovskites is discussed and organized according to structural engineering including A-site, B-site substitution, composite engineering, and morphology

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https://doi.org/10.1016/j.ijhydene.2022.09.130
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Rare-earth metals
Substitution
Structural engineering

engineering. At last, the challenges and opportunities confronting rare-earth-based perovskites catalysts in superior OER are discussed.

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Introduction

The international community has promoted vigorously the development of sustainable energy sources due to the energy and environmental crisis originating from the huge consumption of conventional fossil fuels. It is expected to improve the share of clean energy which effectively meets the growing energy needs and low-carbon economy [1–3]. Hydrogen generated from the electrochemical water splitting is an ideal clean energy source [4–6]. Electrochemical water splitting composes of hydrogen evolution reaction (HER) at the cathode and oxygen evolution reaction (OER) at the anode [7–9]. Due to the complex four-electron transfer mechanism of OER ($4OH^- \rightarrow O_2 + 2H_2O + 4e^-$), its kinetics is relatively slow and considered the rate-determined step [10,11]. Ir and Ru-based catalysis perform excellently for OER [12,13], however, the high expense and low reserves of these precious metals hinder their commercialization. And thus, developing low-cost, abundant, and highly active OER catalysts is of great significance [14–16].

The activity and stability of non-precious transition metals and their compounds in alkaline solutions for OER have been universally investigated recently, such as oxides of 3*d* transition metals [17–19], sulfides [20], phosphides [21], perovskites [22,23], and hydro/oxides [24,25] to take the place of precious metal-based electrocatalysts. Due to the low cost, an abundance of earth reserves and coordinated electronic structures, perovskite has attracted great interest and has been universally applied in solar cells [26–28], electrocatalytic water decomposition [29,30], metal-air cells [31,32], luminescent materials [33,34], and temperature sensing [35]. More importantly, some rare-metal-based perovskites have catalytic capabilities similar to or higher than those of precious metals-based electrocatalysts [36]. For instance, Li et al. [37] proposed a technique by modulating the atomic percentage of iron and nickel in the perovskite oxide of $LaFe_{0.25}Ni_{0.75}O_3$ and found that the further selenium-doped $LaFe_{0.25}Ni_{0.75}O_3$ present excellent OER performance with a low overpotential of 287 mV to afford the current density of 10 mA cm^{-2} and a small Tafel slope of 87 mV dec^{-1} in alkaline electrolyte, distinctly superior to that of the present perovskite oxide and the commercial RuO_2.

Rare-metal-based perovskite has the typical formula of ABO_3, where the A-site comprises rare-earth metals and B-site is a transition metal. The coordination number of the A-site element is 12, such as AO_{12}, whereas the coordination number of the B-site element is 6, such as BO_6. In addition, there are different crystal structure derivatives of perovskites, including single perovskites (ABO_3) [38,39], dual-perovskites ($A_2B_2O_6$) [31], Ruddlesden-Popper perovskites ($A_{n+1}B_nO_{3n+1}$, (n = 1, 2, and 3)) [40]. The physical, chemical and electrocatalytic behaviors could be modified by A-site/B-site substitution and composite engineering. This partial substitution would optimize the surface structure as well as tune the electronic configuration of the perovskite, which is the benefit of enhancing their intrinsic catalytic activity and deeper understanding of the relationship between the structure and electrocatalytic performance. Therefore, various strategies have been adopted to modulate the electronic configuration of rare-earth-based perovskites to enhance their catalytic capability for OER. For instance, Mefford et al. [41] proposed the controlling of the covalent nature

of bonds and the content of O-vacancy by Sr^{2+} substituting La^{3+} of $La_{1-x}Sr_xCoO_{3-\delta}$ perovskite, thereby revealing the electronic structure and giving rise to the higher activity. The stoichiometry of O in the perovskite crystals often varies from the nominal value of 3 to form the ABO_3, tuning both the unstable oxygen status at the surface as well as the underlying electronic configuration of the electrocatalysts. The amount of O-vacancy in the perovskites is associated with the relative positions of the 3*d* bands of the transition metals to the 2*p* band of the O in the crystal. The more the covalent systems, the higher the vacancy concentration.

In this review, the crystal structure and synthesis techniques of rare-earth-based perovskites are discussed and the recent advance in perovskites-based catalysts is reviewed from the viewpoint of OER. Importantly, the structure engineering strategies including A-site, B-site substitution, composite engineering, and morphology engineering are discussed systematically (Fig. 1) and the challenges and prospects of perovskites-based OER catalysts are addressed.

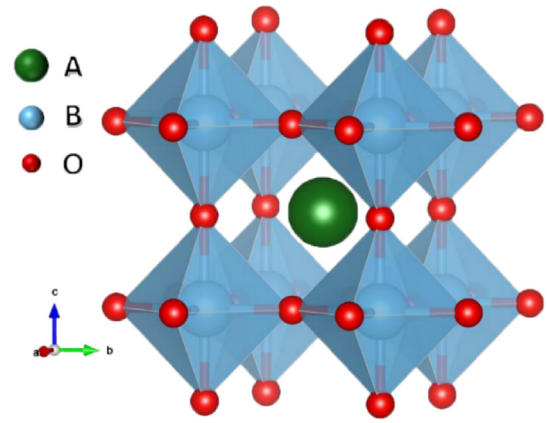

Fig. 2 – Crystal structure of ABO_3 perovskite. Reproduced with permission [45] and copyrighted 2021, HUSCAP.

Structure and properties of rare-earth-based perovskites

The ideal crystal structure of rare-metal-based perovskite is cubic and associated with space group Pm-3m, as shown in Fig. 2. The cavities of the BO_6 octahedral framework in the perovskite structure are filled with cations. The inclination angle of B–O–B is between 140° - 180°. The small variation in bond length and bond angle of the crystal structure significantly regulate the physical properties. It has been reported that the perovskites show various outstanding physical properties, such as high electrical conductivity [42], magnetic order [43], ferroelectricity [44], etc.

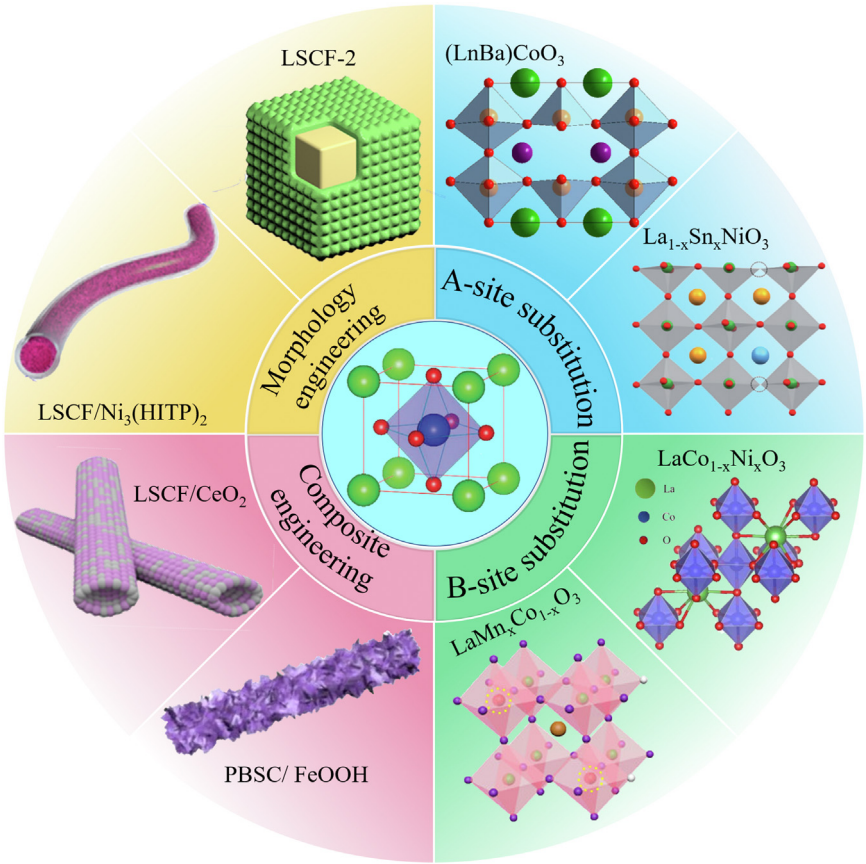

Fig. 1 – Schematic illustration of the structural engineering of perovskites. Copyright permission has been received for all the images.

Toda et al. [42] reported the layered-structure perovskite oxide $NaLnTiO_4$ (Ln = rare-earth-based elements) and analyzed the effect of crystal structure on electrical conductivity. The compounds with different rare-earth-based elements at the A′ position show different symmetries. When Ln = La–Nd, the compounds show tetragonal symmetry, whereas the Ln = Sm–Lu shows orthogonal symmetry. At high temperatures, the frail interaction between the perovskite oxides layer and interlayer Na ions leads to much higher ionic conductivity of single perovskite layer $NaLaTiO_4$ than that of triple perovskite layer $Na_2La_2Ti_3O_{10}$. Ovchinnikov et al. [46] found that the disordered states of gadolinium and strontium in the lattice of the $Gd_{0.2}Sr_{0.8}CoO_{3-\delta}$ perovskite are significant to the physical properties. The larger radius of Sr_{2+} lead to local deformation and the disorder of the Sr^{2+}/Gd^{3+} ions at the A site of the lattice gives rise to higher catalytic activity and selectivity than those of the ordered state.

Not only the above physical properties but also the changes in the crystal structure of perovskite oxides affect their catalytic capability in OER. For instance, Lee et al. [47] reported that the strain on the crystal structure of $LaNiO_3$ perovskite will result in an enhancement in the electrocatalytic performance. A compressive strain of 1.2% could improve the OER performance which is higher than the best performing noble metal catalysts, ascribing to the strain-induced shift of the *d*-band center.

The catalytic mechanism of perovskite oxides is explained according to the bonding orbital and antibonding orbital interaction of perovskite oxides: the interaction between the σ bond and O is larger than the t_{2g} orbital of the π bond, which could more directly enhance the electrons between the cations at the surface and the adsorbed intermediates (*OH, *OOH, *OO) [36]. Additionally, in alkaline media, the perovskite catalyst undergoes surface reconstruction to generate new species, such as MOOH (M represents metal), which normally presents superior electrocatalytic capability than that of the host materials to serve as the real electroactive centers for OER [48–51]. To understand the surface revolution and reconstruction during the OER, *in situ* techniques should be applied, which are important to reveal the catalyst evolution during the reaction [52–54]. For instance, *in situ* XRD (X-ray diffraction) can be used to detect the crystal structure change (crystal or amorphous, crystal form, lattice parameters, etc.) of the rare-earth-based perovskite catalysts during the reaction. The *in situ* TEM (transmission electron microscopy) is beneficial in providing the morphology and crystal lattice change in the OER process.

Synthesis of rare-earth-based perovskites

Various techniques including the sol-gel method, hydrothermal/solvothermal method, electrostatic spinning method, pulsed laser deposition and so on have been normally applied to synthesize the perovskites electrocatalysts. The effect of the preparation methods on the crystal shape, surface morphology, as well as electrocatalytic mechanism for OER has been addressed.

Sol-gel method

The sol-gel technique is one of the most commonly adopted methods to synthesize perovskite oxides. Compared with traditional solid-phase reactions, this technique can accurately control the composition of the product, improve the crystallinity of the final structure, and reduce the calcination temperature required. However, the reaction normally takes a long period. The sol-gel process typically consisted of the following steps: (i) mixing the raw material and stirring to form sol, (ii) static ageing to form a gel, (iii) drying to form a dry gel, and (iv) annealing to obtain perovskite oxide.

The sol-gel method commonly generates particles with large agglomeration. However, nanoscale products can also be prepared through subsequent processing. For example, Gao et al. [55] synthesize $LaMn_xCo_{1-x}O_3$ (x = 0, 0.2, 0.4, 0.6, 0.8 and 1) samples by the sol-gel technique. The precursor is obtained by the sol-gel technique and annealing in air at 600 °C for 3h. With the increase of Mn doping content, the diffraction peaks between 32° and 34° shift to higher angles, agreeing well with the increase of the covalent radius induced by the substitution of Co by Mn. He et al. [56] reported that $LaMn_{1-x}Co_xO_3$ perovskite catalysts with different Co contents have been synthesized by the sol-gel protocol, as shown in Fig. 3. The solution of lanthanum nitrate, cobalt nitrate and manganese nitrate is prepared according to the stoichiometric ratio. Moreover, the preparation method previously used was improved by adding citrate as a complexing agent to accelerate the reaction [57–60].

Yoon et al. [61] proposed that the $LaNi_{0.6}Fe_{0.4}O_3$-CeO_2 (LNF-C) perovskite composite was prepared by a citrate-assisted

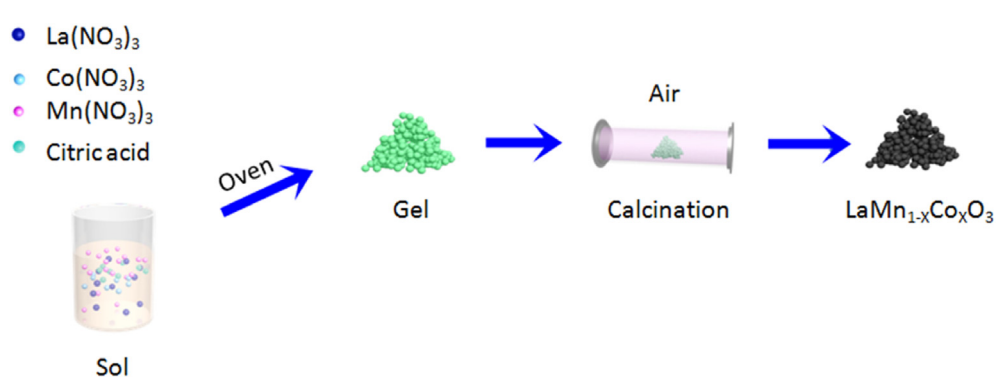

Fig. 3 – Scheme for fabrication of $LaMn_{1-x}Co_xO_3$. Reproduced with permission [56] and copyrighted 2018, Wiley-VCH.

sol-gel technique. Firstly, a certain proportion of lanthanum salt, nickel salt, iron salt, cerium salt, and citrate are mixed and stirred to prepare a sol. Then, the pH of the mixture is tuned to 7–8 and kept at 80-90 °C for 4 h and another 5 h at 200 °C to prepare a dry gel. Finally, the dry gel was annealed at 950 °C for 5 h to prepare LNF-C powder. Gui et al. [62] reported that PBFN2.5 has been synthesized *via* an EDTA-citrate-assisted sol-gel technique, the precursor was synthesized by the sol-gel protocol, and the product was obtained by high-temperature annealing. EDTA is added based on the standard sol-gel method, followed by tuning the pH of the mixture to neutral, thereby obtaining submicron particle size and aggregates to form granular aggregates.

Hydrothermal/solvothermal method

Hydrothermal/solvothermal method is a simple wet chemical synthesis method, where chemical reactions occur in solutions at high temperatures and pressure to precipitate crystalline products. And thus there is a security threat as well as a high requirement for the equipment due to the high temperature and high pressure. The rare-earth products usually require a combination of hydrothermal/solvothermal methods with several other synthetic methods. Nevertheless, compared to the sol-gel synthesis, the following annealing temperature required for the hydrothermal is significantly lower. Furthermore, hydrothermal synthesis realizes the formation of products with different morphologies, such as nanospheres, nanoflowers, nanosheets, nanorods, and core-shell structures.

Rezanezhad et al. [63] prepared the $LaFeO_3$ perovskite electrocatalysts using a one-pot hydrothermal reaction. Firstly, ferric nitrate, lanthanum nitrate, and manganese nitrate with a certain proportion were dissolved in water and stirred for 15 min. Then, the pH was adjusted to 12 with ammonia water, and afterwards, the hydrothermal protocol was conducted at 160 °C for 24 h. Finally, annealing in air at 800 °C for 4 h to obtain $LaFe_{1-x}Mn_xO_3$. Chu et al. [64] prepared $LaCo_xNi_{1-x}O_3$ nanorods by hydrothermal reaction at 180 °C for 10 h and followed by annealing in a tube furnace, as schematically illustrated in Fig. 4.

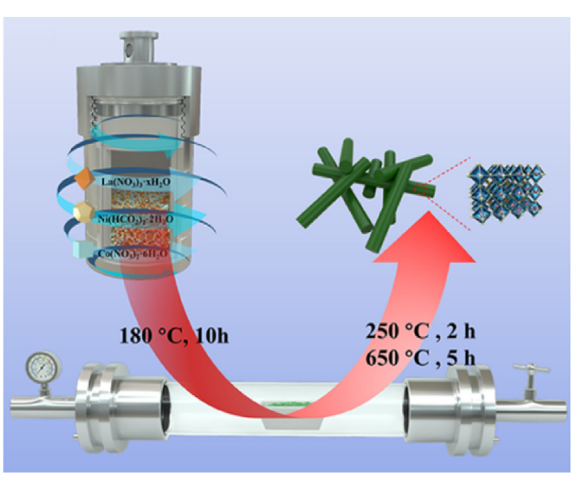

Fig. 4 – Schematic illustration of the synthetic procedure for dual-perovskite $LaCo_xNi_{1-x}O_3$ (LCNO) nanorods. Reproduced with permission [64] and copyrighted 2021, American Chemical Society.

Compared with traditional hydrothermal reactions, the use of microwave and ultrasonic waves in chemical synthesis can effectively increase chemical yield, improve purity, and shorten reaction time. Chen et al. [65] prepared the $La_{1-x}Ce_xCoO_3$ perovskite electrocatalysts using microwave- and ultrasonic-assisted hydrothermal reactions. Lanthanum nitrate, cerium nitrate, and cobalt nitrate are mixed according to a certain stoichiometric ratio, and then the pH value is adjusted to 12 with NaOH. The hydrothermal reaction is carried out with a microwave reactor for 210 °C for 1h and finally, the sample was annealed at 500 °C for 2 h to produce black perovskite powder. This method transfers energy rapidly and directly to the reactants, the internal and external temperatures are raised simultaneously, and the nanocrystalline perovskite catalysts prepared by microwave and ultrasonic waves can rapidly nucleate and grow.

Electrospinning method

Electrospinning is considered an easy, cheap, and versatile method to prepare one-dimensional nanomaterials with tunable composite and morphology. Recently, numerous researchers have investigated the synthesis of one-dimensional rare-earth doped materials and chalcogenide structures, such as nanowires, nanotubes, etc. The left part of Fig. 5 is a diagram of the electrospinning device, which consists of a power supply to provide high-voltage output, an injector with a metal needle, and a grounded conductive collector. The electrospinning synthesis process is mainly divided into three steps: (i) preparing the initial solution, (ii) electrospinning of nanofibers, and (iii) calcining of the nanofibers to obtain the product. In this preparation protocol, the organic solvent in the spinning solution is toxic and volatile, which needs a high cost to remove.

Li et al. [37] synthesize the $LaFe_{1-x}Ni_xO_3$ perovskite catalyst by electrospinning. The metal salt was dissolved in 10 mL of DMF. PVP (15 wt%) powder was then added and stirred overnight. Transfer the resulting precursor solution into a plastic syringe for electrospinning. The solution feeding rate was fixed at 0.6 mL h^{-1}, and the working voltage was fixed at 18 kV. Finally, the sample was undergone high-temperature annealing. The nanofiber/nanorod structures prepared by electrostatic spinning can obtain a large specific surface area, which serves as excellent catalyst precursors and is followed by modification with other functional materials to enhance the catalytic activity. Zhang et al. [66] used the same procedure to prepare $PrBa_{0.5}Sr_{0.5}Co_2O_{5+\delta}$ nanorods, as illustrated in Fig. 5. During the electrospinning process, the relative humidity was adjusted to be 30%–35%, and the voltage and the working distance were applied. Then, annealed at 800 °C for 5 h to obtain the primary product. In the end, the PBSC@FeOOH core-shell structure was prepared by atomic layer deposition.

Pulsed laser deposition

Pulsed laser deposition (PLD) is a method for the preparation of multi-component stoichiometric materials (especially

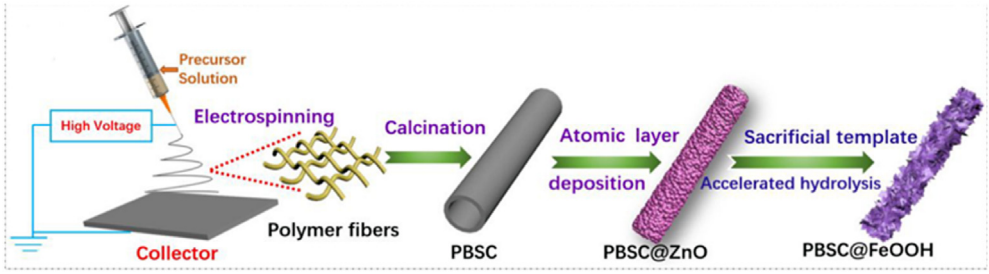

Fig. 5 – Fabrication of core-shell PBSC@FeOOH-20 catalysts. Reproduced with permission [66] and copyrighted 2018, American Chemical Society.

perovskite oxides) and the controllable preparation of thin-film materials with a short deposition time. In recent years, with the development of laser technology, PLD technique has attracted more and more attention although it is not applicable for mass production. The preparation of thin-film materials by this technique is generally divided into four stages: i) the interaction of the laser radiation and the target; ii) the dynamics of the melt; iii) the deposition of the melt on the substrate; iv) the generation of the film nuclei at the surface of the substrate and growth. There are small molten particles in the film, which will greatly reduce the quality of the film. Factors affecting film growth throughout the preparation process include target type, pulsed excimer laser (laser density and laser repetition rate), substrate type, the temperature at the substrate, the distance between target and substrate, oxygen partial pressure, etc. [67–70] For instance, Pan et al. [68] used PLD technique to prepare $LaCoO_{3-\delta}$ nanoparticles grown on nickel foam, as shown in Fig. 6. When preparing the $LaCoO_{3-\delta}$/NF thin-film material, the substrate was pretreated at first, and then the parameters were controlled. The temperature of the substrate was set to 750 °C, the oxygen partial pressure in the cavity was 100 mTorr, the pulse repetition frequency was 10 Hz, and the ablation flux was 0.4 J cm^{-2}.

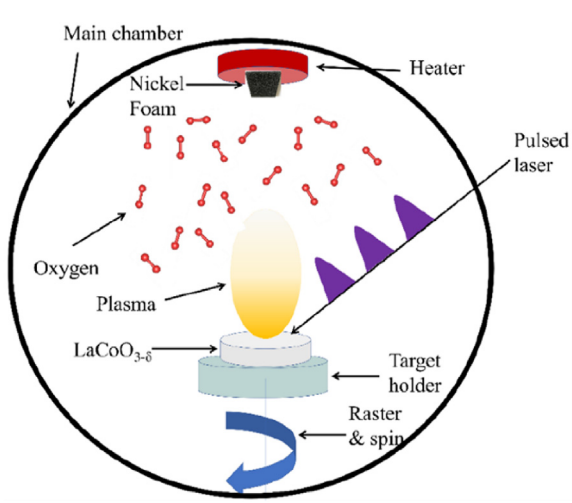

Fig. 6 – Scheme of the preparation of $LaCoO_{3-\delta}$ on nickel foam *via* PLD technique. Reproduced with permission [68] and copyrighted 2022, Elsevier.

Hu et al. [71] used PLD technology to grow $NdNiO_3$ thin films on single crystal $SrTiO_3$ (STO) (001) substrates (5 × 10 mm^2). The $NdNiO_3$ target should be prepared by traditional high-temperature solid-phase synthesis firstly, and then followed by depositing on the STO (001) substrate using PLD technology. During the preparation process, the oxygen vacancy concentration of the material is modulated by controlling the oxygen partial pressure, thereby affecting the surface structure of the material. Andrei et al. [72] reported that the nanostructured $LaFeO_3$ has been prepared on Nb:$SrTiO_3$ (STON) substrate by PLD technique using perovskite oxide $LaFeO_3$ as a ceramic target. The thickness of the thin film is regulated by the number of laser pulses, and the thickness of the prepared film varies from 14 to 200 nm.

Other methods

In addition to the above preparation processes, high-temperature solid-phase synthesis is also a simple synthesis. However, the reaction temperature is as high as 1000 °C, which consumes too much electrical energy, resulting in poor economic efficiency. There are also some more complex synthesis processes, such as solution combustion method, magnetron sputtering method, stencil method, atomic layer deposition etc. For example, Kim et al. [73] prepared the $La_{0.5}Sr_{0.5}MnO_3$ powder by sol-gel method firstly, followed by depositing nano-scale cobalt oxide (Co_3O_4) *via* atomic layer deposition technology.

Structural engineering

Generally, the pristine perovskites perform moderately in OER due to the unsatisfied intrinsic activity. Regulating the structure of rare-earth-based perovskites is considered a useful route to improve the electrocatalytic performance of OER in the alkaline solution [36]. The perovskite crystal configuration belongs to the face-centered cubic structure. The A site occupies the cube vertex, the B site occupies the body center, and the oxygen element occupies the face center position. When the heteroatoms enter the structure, the perovskite lattice is distorted, more active sites are exposed, and the intrinsic activity of the catalyst is improved; compound modification increases the catalytic energy of the catalyst through synergy. As shown in Table 1, the structure regulation methods can be divided into three categories including A/B-

site substitution, composite engineering, and morphology engineering.

A-site substitution

The cations at the A-site in a perovskite oxide can tune the valence state, crystal structure, electrochemical active area, and oxygen vacancies of B-site ions, such as $LaFeO_3$ [76], $LaCoO_3$, and $LaNiO_3$ [77]. The ions of different radii replace the A-site ions to regulate the crystal structure, electronic configuration, and surface chemical behaviors. The substitution of transition metals at the A-site represents the enhancement of the electrochemical performance of perovskite electrocatalysts by modulating the electronic arrangement. For example, Liu et al. [97] proposed an effective strategy to replace the A-site cations with Sn to enhance the catalytic activity of $LaNiO_3$ for OER, as shown in Fig. 7a. When part of La ions are replaced by Sn ions, the diffraction peaks gradually become larger, indicating the lattice shrinkage by the introduction of Sn. The enhanced electrocatalytic activity of $La_{1-x}Sn_xNiO_{3-\delta}$ perovskite is attributed to Sn substitution and the introduction of O-vacancy at the surface. To generate a current density of 10 mA cm^{-2}, the overpotential of $La_{0.9}Sn_{0.1}NiO_{3-\delta}$ (LSN-0.1) is 318 mV and the catalyst works stably for more than 20 h, as well as the C_{dl} of LSN-0.1 is 42.07 mF cm^{-2}, as shown in Fig. 7b–c. Under Sn substitution, the interfacial charge transfer of LSN-0.1 is faster and exhibits improved OER activity and stability at room temperature comparing to the pristine perovskite.

Controlling the selective substitution at the A site leads to the dynamic reconfiguration of the perovskite structure, thereby enhancing the OER performance. Sun et al. [98] proposed an improved perovskite with A-site selective substitution that accelerates the surface reconstruction process of the catalyst surface, thereby enhancing OER catalytic capability. Doping with Ce into the A-site precisely reduces the reconfiguration potential of the electroactive phase and establishes dynamic structure-activity correlations. The density-functional theory calculations show that the Ce substituted A-site increases oxygen 2*p* levels and optimizes the concentration of O-vacancy, thus enabling the B-site atoms, and enhancing the electroactive phase remodelling. As shown in Fig. 7d, increasing the concentration of Ce dopant, the reconfiguration potential decreases obviously, and when the doping amount exceeds 0.1, the applied voltage drops by 100 mV. With the increase of O-vacancy content, the flexibility

Table 1 – OER activity of different structures of rare-earth-based catalysts.

Methods	Electrocatalyst	Overpotential at 10 mA cm^{-2} [mV]	Electrolyte	Reference
A-site substitution				
	$La_{0.2}Sr_{0.8}FeO_3$	318 at 50 μA cm^{-2}	1.0 M KOH	[74]
	$Pr_{0.5}Ba_{0.3}Ca_{0.2}CoO_{3-\delta}$	440	0.1 M KOH	[75]
	$La_{0.96}Ce_{0.04}CoO_{3-\delta}$	380	1.0 M KOH	[65]
	$La_{0.95}FeO_3$	90	0.1 M KOH	[76]
	$La_{0.95}NiO_3$	400	1.0 M NaOH	[77]
	$GdBa_{0.6}Sr_{0.4}Co_2O_{6-\delta}$	70	0.1 M KOH	[78]
	$PrBa_{0.85}Ca_{0.15}MnFeO_{5+\delta}$	400	0.1 M KOH	[79]
B-site substitution				
	$La_{0.2}Sr_{0.8}Ni_{0.8}Fe_{0.2}O_3$	430 mV at 217 μA cm^{-2}	0.1 M KOH	[80]
	$LaMn_{0.7}Co_{0.3}O_3$	220	0.1 M KOH	[56]
	$La_{0.5}Sr_{0.5}Ni_{0.4}Fe_{0.6}O_{3-\delta}$	770 at 26.4 mA cm^{-2}	0.1 M KOH	[4]
	$GdFe_{0.8}Cu_{0.2}O_3$	660 at 9.66 mA cm^{-2}	1.0 M KOH	[81]
	P-$LaFeO_{3-\delta}$	465	0.1 M KOH	[82]
	$PrBaCoFeO_{6-\delta}$	50	0.1 M KOH	[83]
	$LaNi_{0.8}Fe_xCu_{0.2-x}O_{3-\delta}$(x = 0.2)	770 at 8 mA cm^{-2}	0.1 M KOH	[84]
	$NdBa_{0.75}Ca_{0.25}Co_2O_{5+\delta}$	14	0.1 M KOH	[22]
	$LaCo_{0.9}Fe_{0.1}O_3$	400 at 0.272 mA cm^{-2}	1.0 M KOH	[85]
	$LaCo_{0.8}V_{0.2}O_3$	306	1.0 M KOH	[86]
	$LaMn_{0.3}Co_{0.7}O_3$	416	0.1 M KOH	[87]
	$PrBa_{0.5}Sr_{0.5}Co_{1.5}Fe_{0.5}O_{5+\delta}$	300	0.1 M KOH	[88]
	$Pr_{0.5}Ba_{0.5}Fe_{0.975}Ni_{0.025}O_{3-\delta}$	270	0.1 M KOH	[62]
Composite engineering				
	$La_{0.8}Sr_{0.2}MnO_{3-\delta}/Ba_{0.5}Sr_{0.5}Co_{0.8}Fe_{0.2}O_{3-\delta}$	380	0.1 M KOH	[89]
	IrO_2-$La_{0.8}Sr_{0.2}MnO_3$	15	0.1 M KOH	[90]
	CoP-$PrBa_{0.5}Sr_{0.5}Co_{1.5}Fe_{0.5}O_{5+\delta}$	440	1.0 M KOH	[91]
	FeOOH-$PrBa_{0.5}Sr_{0.5}Co_2O_{5+\delta}$	380	0.1 M KOH	[66]
	Co-Pi- $La_{0.7}Sr_{0.3}MnO_3$	220	1.0 M NaOH	[92]
	$LaNiO_3$@FeOOH	264	1.0 M KOH	[104]
Morphology engineering				
	Hollow and porous $La_{0.7}Sr_{0.3}Co_{0.25}Mn_{0.75}O_3$	340	1.0 M KOH	[93]
	3D ordered microporous $LaFeO_3$	410	1.0 M KOH	[60]
	Heterogenous $LaNiO_3$-NCNTs	390	0.1 M KOH	[94]
	$La_{0.4}Sr_{0.6}Co_{0.7}Fe_{0.2}Nb_{0.1}O_{3-\delta}$ porous	360	0.1 M KOH	[95]
	Amorphous layer of $La_{0.8}Sr_{0.2}Co_{0.8}Fe_{0.2}O_{3-\delta}$ nanoparticles	248	1.0 M KOH	[96]

of the electrocatalyst is enhanced, and the OER capability of $La_{0.9}Ce_{0.1}NiO_3$ is significantly improved. In addition, the $La_{0.9}Ce_{0.1}NiO_3$ catalyst remained stable for 48 h, which is crucial for an electrocatalyst. Fig. 7e shows that when the Ce is partially substituted, the hybridization of Ni_{3d}-O_{2p} shows a significant enhancement, suggesting that the Ni-O orbital hybridization weakens with the Ce dopant increasing. When the amount of Ce substitution is greater than 0.1, the weakening of Ni-O hybridization hinders the redox of oxygen, and more energy barriers need to be overcome to achieve the phase transition. Substitution at A site will not only produce corresponding defects, but also lattice distortion, site reconstruction, and valence changes of B site elements.

Zhu et al. [76] introduced A-site cation defects into $LaFeO_3$ perovskite, as shown in Fig. 7f, the generation of O-vacancy at the surface of the materials and the formation of a small

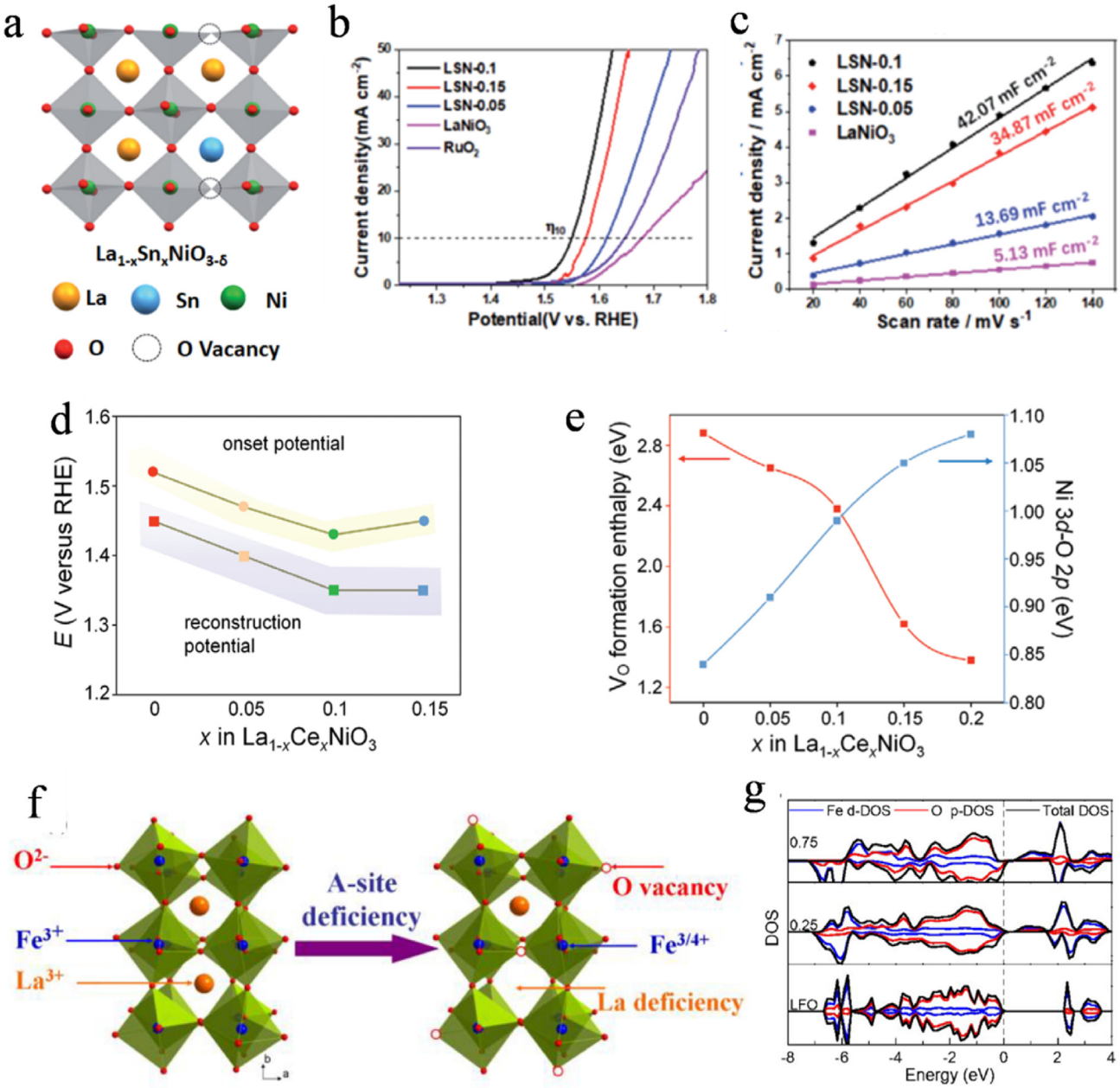

Fig. 7 – (a) The crystal model of $La_{1-x}Sn_xNiO_{3-\delta}$. (b) OER polarization curves. (c) The electrochemically active surface area (ECSA) results for $La_{1-x}Sn_xNiO_{3-\delta}$ and RuO_2 under the alkaline condition. Reproduced with permission [97] and copyrighted 2022, the Royal Society of Chemistry. (d) The relationship between surface reconstruction potential and the corresponding onset potential for OER. (e) Calculated O-vacancy (V_O) formation enthalpy and the Ni_{3d}-O_{2p} PDOS band center of the cells for $La_{1-x}Ce_xNiO_3$ (x = 0, 0.05, 0.1, 0.15, and 0.2). Reproduced with permission [98] and copyrighted 2021, Wiley-VCH. (f) Scheme of the generation of O-vacancy and Fe(IV) in A-site-deficient La_{1-x} $FeO_{3-\delta}$ perovskites. Reproduced with permission [76] and copyrighted 2016, American Chemical Society. (g) The density of states (DOS) for LFO, LFO-0.25, and LFO-0.75. Reproduced with permission [74] and copyrighted 2020, the Royal Society of Chemistry.

amount of Fe(IV) enhanced the OER catalytic activity. There are two reasons: (i) the surface of $La_{1-x}FeO_{3-\delta}$ perovskite lacking A-site cations forms oxygen vacancies reflected by a high oxidation state (O_2^{2-}/O^-), and (ii) little Fe(IV) species (t_{2g}^3 e_g^1) with optimal e_g orbital filling is present in $La_{1-x}FeO_{3-\delta}$- perovskites. Due to the partial absence of the cation at the A-site, the chemical valence of the ion at the B-site increases. On this basis, Shen et al. [74] studied the effect of A-site Sr substitution of La on the electrocatalytic performance of perovskite $LaFeO_3$ (LFO). The partial substitution of La by Sr in $LaFeO_3$ resulted in the conversion of Fe(III) to Fe(IV), and the OER capability was distinctly enhanced. The presence of Fe(IV) makes the Fe_{3d}-O_{2p} orbital hybridization of the Fe-O orbital stronger, which shifts the valence band energy position (VB) towards the Fermi level. This electronic regulation modifies the adsorption energy of the *OH intermediate at the surface, contributes to the accelerated OER process, and generates a new hole at ~0.9 eV below the conduction band, bringing the electron transfer energy barrier from 1.34 to 0.4 eV to facilitate the charge transport at the interface. And the Fe(IV)-induced electronic state has better OER performance. To systematically investigated the influence of Sr substitution on the electronic structure of LFO, DFT+U (U = 3 eV) calculation method was applied to study the electronic structure of LSFO. Sr replaces La to generate strongly hybrid hole states of Fe_{3d} and O_{2p} (Fig. 7g). The overlap of O_{2p} and Fe_{3d} partial density of states (PDOS) in both occupied and unoccupied states increases with the increase of Sr content, which enhances the hybridization of Fe_{3d}-O_{2p} by the higher Fe oxidation state. In addition, the LFO-0.8 electrocatalyst presents excellent stability that it can work for more than 24 h without obvious decay.

B-site substitution

The B-site cations substitution of perovskites with high-valence cations of different sizes can improve the coupling efficiency of each pair of B-site atoms in the intercalated oxygen bridge, which can tune their electronic structure and charge transport properties and optimize OER performance to understand OER activity [36]. Li et al. [86] prepared the V-doped $LaCoO_3$ catalysts and found that the V-doping enhances the intermediates' adsorption in OER. Moreover, the amorphous shell formed by vibrational exchange interactions between high-spin V(IV) and low-spin Co(II) with appropriate thickness can promote catalytic activity.

Fe doping is beneficial to the formation of Co^{4+} and oxygen ligand pores, which are highly active in the formation of highly oxidized oxygen [99]. In addition, the doping ionic radius also significantly affects the crystal structure and catalytic activity. Wang et al. [100] discussed the effect of the Ni:Co ratio on perovskite crystal structure and catalytic activity. The radius of Ni^{3+} ionic is larger than that of the Co^{3+}, as a result, the substitution of cobalt with nickel leads to the expansion of crystal lattice, and the diffraction peak shifts to lower angles with increasing Ni substitution. The enhanced catalytic capability may originate from the synergistic effect of Co and Ni existing in mixed-valence states of Co^{3+}/Co^{2+} and Ni^{3+}/Ni^{2+} with modified composition and regulable ability to adsorb *OH and desorb O_2, as illustrated in Fig. 8a–b.

Gao et al. [55] proposed that the substitution of Mn for Co sites was found to improve the OER electrocatalytic performance of $LaCoO_3$. The incorporation of Mn could enlarge the concentration of Mn(IV) and Co(III) which are highly electroactive. The relative strength of Co^{3+}/Co^{2+} is high, and the optimal doping content of Mn can improve the electrical conductivity and intrinsic electroactivity of the catalyst, and thus enhance the electrocatalytic performance. As shown in Fig. 8c, $LaMn_{0.4}Co_{0.6}O_3$ presented the outstanding OER performance among the $LaMn_xCo_{1-x}O_3$ electrocatalysts. After Mn doping, the density of states (DOS) at the Fermi level (E_F) is in a strong hybrid occupied state, and the Co 3*d* orbitals pass through E_F, showing metal-like characteristics. Fig. 8d–e suggested that the doped Mn would improve the conductivity of $LaCoO_3$ and accelerate the charge transfer rate during OER. In addition, $LaMn_{0.4}Co_{0.6}O_3$ exhibits remarkable OER persistence, which remains stable for 80,000 s at 1.6 V *vs*. RHE.

Perovskites have large numbers of electroactive centers and are highly stable, and modulating the electronic structure could result in better OER catalytic performance. Wu et al. [10] proposed that the vibronic super-exchange of Ni^{3+}-O-Mn^{3+} in La_2NiMnO_6 nanoparticles induced the modified e_g electron filling state of Mn and Ni ions toward unity. The strong Jahn-Teller distortion of MnO_6 and NiO_6 octahedra was formed by the vibronic super-exchange interaction, which attributed to the generation of the electroactive species of Mn/Ni hydroxides/oxides at the surface of the electrocatalysts, as shown in Fig. 8f. Therefore, La_2NiMnO_6 nanoparticles exhibited excellent OER catalytic performance.

Composite engineering

Composite engineering is universally investigated to enhance the OER capability of pristine perovskite oxides catalysts due to their limited intrinsic activity. Gong et al. [66] introduced a core-shell structured $PrBa_{0.5}Sr_{0.5}Co_2O_{5+\delta}$ (PBSC)@FeOOH catalyst, as shown in Fig. 9a, where loose flakes of FeOOH uniformly cover the smooth surface of PBSC. The FeOOH covering the surface of PBSC increases the specific surface area and forms a porous structure, which increases its catalytic activity. Strong electronic interactions between PBSC and FeOOH modulate the interactions between the intermediates and the catalyst. The amorphous FeOOH layer on the catalyst not only forms a dense protective film but also increases the catalytic activity. Not only the super catalytic capability but also the outstanding long-term stability of the catalyst is achieved. After 500 consecutive cyclic scans of the PBSC@FeOOH-20, the current loss is small, demonstrating the good stability of PBSC@FeOOH-20.

Bhowmick et al. [92] explored the improved OER activity of Co-Pi@$La_{0.7}Sr_{0.3}MnO_3$ catalysts. In Sr-doped $LaMnO_3$, the introduction of Sr^{2+} into the La^{3+} lattice led to the formation of Mn^{4+} from Mn^{3+}, inducing the formation of cavities in the lattice. In addition, the doping concentration of 30% gives a good electrocatalytic performance with good stability that the catalyst is stable after 16 h of continuous operation under 1.45 V *vs*. RHE. The Co^{2+} at the surface of $La_{0.7}Sr_{0.3}MnO_3$ is oxidized to Co^{3+}/Co^{4+} and moves to the electrolyte interface, and then interact with H_2O molecules to release the holes, as a result, the O_2 generated and reduces itself to Co^{2+}. The rapid

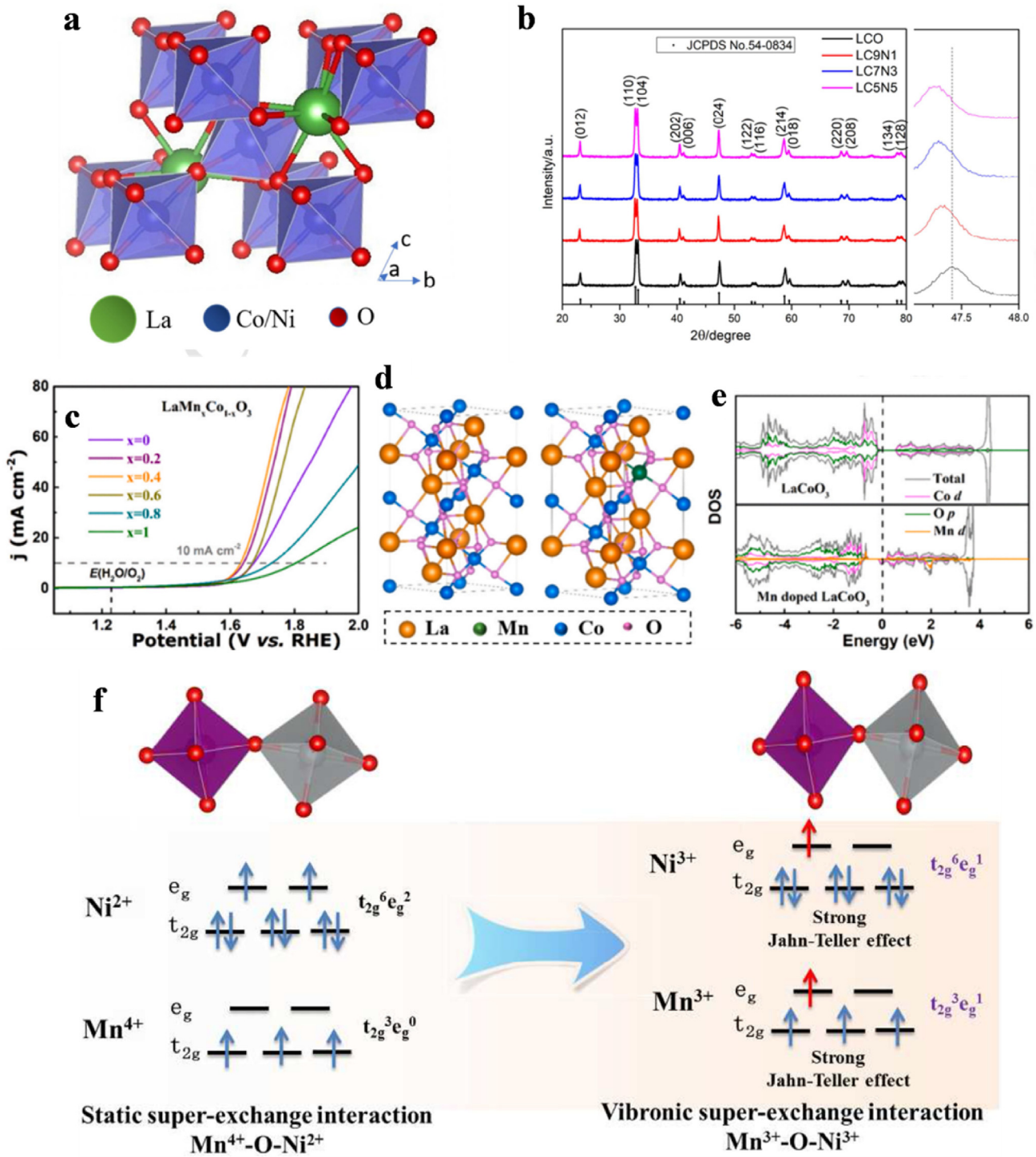

Fig. 8 – (a) Scheme of the crystal structure of $LaCo_{1-x}Ni_xO_3$. (b) XRD patterns for $LaCo_{1-x}Ni_xO_3$. Reproduced with permission [100] and copyrighted 2019, Elsevier. (c) Polarization curves of $LaMn_xCo_{1-x}O_3$. (d) The crystal structure and (e) DOS diagrams of $LaCoO_3$, Mn-doped $LaCoO_3$. Reproduced with permission [55] and copyrighted 2021, Elsevier. (f) The scheme for the deformation of MnO_6 and NiO_6 octahedron. Reproduced with permission [10] and copyrighted 2018, American Chemical Society.

shuttling of the oxidized state of the Co ion between +2 and + 3/+4 gives rise to faster and more efficient OER kinetics. Liu et al. [104] investigated that $LaNiO_3$@FeOOH enhances OER performance. The excellent OER activity was obtained from the adsorption energy and facilitated charge transport of oxygen-containing substances. FeOOH has strong adsorption energy for oxygen-containing intermediates (*O, *OH, *OOH) at the active sites as surface modifiers. In Fig. 9b, the positive shift of

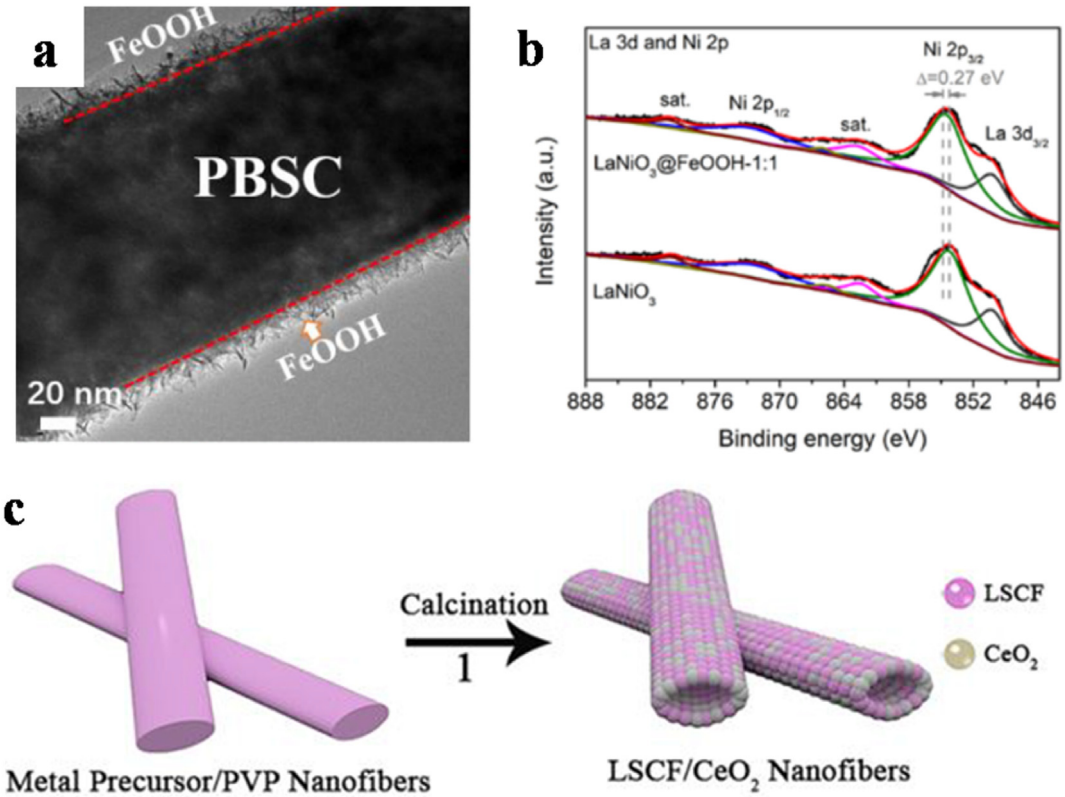

Fig. 9 – (a) TEM images of PBSC@FeOOH-20. Reproduced with permission [66] and copyrighted 2018, American Chemical Society. (b) High-resolution XPS spectra: La 3*d* and Ni 2*p*. Reproduced with permission [104] and copyrighted 2020, Elsevier. (c) Scheme of the LSCF/CeO_2 nanofibers. Reproduced with permission [102] and copyrighted 2019, American Chemical Society.

the peak of Ni $2p_{3/2}$ by 0.27 eV occurs in $LaNiO_3$@FeOOH-1:1 with electron transfer from $LaNiO_3$ to FeOOH, which affects the catalytic activity and the corresponding durability.

To develop a low-cost, high-efficiency electrocatalyst, Wang et al. [90] prepared a composite catalyst of IrO_2-$La_{0.8}Sr_{0.2}MnO_3$. With the synergy between the outstanding oxygen reduction reaction (ORR) performance of $La_{0.8}Sr_{0.2}MnO_3$ and the OER capability of IrO_2 and the enhanced electrochemically active surface area, both the OER and ORR catalytic performance of the composites are enhanced. Jiang et al. [101] constructed RuO_2/LFRO composites from the precipitation solution of low Ru-substituted A-site-deficient perovskite $La_{0.9}Fe_{0.92}Ru_{0.08}O_3$ (LFRO). RuO_2/LFRO exhibits higher OER performance, possibly resulting from the formation of electroactive RuO_2 nanoparticles and enhanced electrical conductivity. Zhang et al. [102] successfully obtained a novel heterostructured composite nanofiber material composed of $La_{0.6}Sr_{0.4}Co_{0.2}Fe_{0.8}O_{3-\delta}$ (LSCF) and CeO_2 nanoparticles by electrospinning method, as shown in Fig. 9c. The nanofibers provide a continuous path for accelerated mass transfer and charge transport, as well as the diffusion of La and Ce at the heterointerface, inducing the generation of O-vacancy. The structure of the fibers suppresses the Sr segregated in LSCF and then improves the durability of the electrocatalyst. The long-term durability of the material was determined by operating at a constant voltage of 0.4 V for 200 h. In the initial test, the LSCF powder had a clear degradation followed by a slight improvement, which was ascribed to the segregation of Sr to the surface or interface.

Morphology engineering

Electrocatalytic reactions occur at the surface of the electrocatalyst, the surface morphology including geometrical shape, particle size, specific surface area, etc. act an important role in catalytic performance. Therefore, the electrocatalytic capability of rare-earth-based perovskite electrocatalysts can be optimized by designing their surface morphology. The preparation and catalytic performance of rare-earth-based perovskite electrocatalysts with unique morphology have been extensively studied as listed in Table 1. For instance, Li et al. [103] proposed the preparation of $La_{0.6}Sr_{0.4}Co_{0.8}Fe_{0.2}O_3$@$Ni_3$(-HITP)$_2$ (LSCF@Ni_3(HITP)$_2$) nanofibers (NFs) by electrospinning, which exhibited strong OER activity and durability. The excellent OER activity depends on the unique nanofibrous structure of the catalyst and the synergistic effect of the coupled conductive layers, which enhance the intrinsic conductivity, accelerated charge transport efficiency, and mass transport capability, thereby enhancing the intrinsic capability of the electrocatalyst. In addition, due to its unique structure, the catalyst has a large specific surface area to improve extrinsic activity, and the high-porous structure strongly adsorbs O-containing adsorbates, stores reaction products, and reduces the adverse effects of O adhesion on electrodes. Thus,

the LSCF@$Ni_3(HITP)_2$ electrocatalyst needs the overpotential of 272 mV to achieve a current density of 10 mA cm^{-2} as well as long-term durability for 12 h in alkaline electrolyte.

The electrospinning technique is useful to prepare hollow and porous nanomaterials. Wan et al. [93] proposed the preparation of hollow and porous $La_{0.7}Sr_{0.3}Co_{0.25}Mn_{0.75}O_3$ nanofibers (LSCM NFs) by electrospinning method and the Ostwald ripening protocol. They found that the hollow $La_{0.7}Sr_{0.3}Co_{0.25}Mn_{0.75}O_3$ nanofibers are composed of a large number of $La_{0.7}Sr_{0.3}Co_{0.25}Mn_{0.75}O_3$ nanoparticles (LSCM NPs). By adjusting the calcination temperature and heating rate, a unique hollow NFs structure is formed, which has a large specific surface area, high porosity, and large internal space, resulting in outstanding OER electrocatalytic capability. The required overpotential of 340 mV is obtained for a current density of 10 mA cm^{-2} and accompanied by long-term stability in a basic electrolyte.

Nanoscale perovskite and amorphous layers with uneven surfaces could supply a large specific surface area, which is beneficial for enhanced electrocatalytic OER. Rational design of the surface morphology of the rare-earth-based perovskite electrocatalysts is crucial to constructing high-performance electrocatalysts. Surface reconstruction of perovskites during OER has been shown to generate the amorphous surface which is beneficial for enhanced OER activity [70,96]. For instance, Zhao et al. [96] proposed that the surface reconstruction of $La_{0.8}Sr_{0.2}Co_{0.8}Fe_{0.2}O_{3-\delta}$ produces an amorphous layer, which increases the specific surface area and improves the wettability of the electrolyte, thereby shortening the electron transport path. The catalyst requires the overpotential of 248 mV (10 mA cm^{-2}) and outstanding stability. Coating the catalyst with carbon material can generate more electroactive centers. Chen et al. [94] proposed the preparation of $LaNiO_3$ nanoparticles in an oil bath, followed by heat treatment to form the composite of nitrogen-doped carbon nanotubes attached to $LaNiO_3$ nanoparticles. The catalyst presents outstanding electrocatalytic capability that the overpotential of 390 mV is needed to afford the current density of 10 mA cm^{-2} and excellent durability, which is associated with the following reasons: i) the large specific surface area of $LaNiO_3$ nanoparticles leads to more electroactive center exposure, ii) $LaNiO_3$ is strongly connected with NCNTs forming a conductive network and promoting charge transfer, iii) the synergistic effect between $LaNiO_3$ and NCNTs.

Conclusion and outlook

In this review, the recent development in the preparation methods and structural engineering including A-site substitution, B-site substitution, and composite engineering of rare-earth perovskite-based compounds as non-precious metal and cheap catalysts for OER are described. Due to the low cost, the abundance of earth reserves, and coordinated electronic structures, perovskite has attracted great interest in electrochemical water splitting. Due to the modified surface structure, modulated electronic configuration and electrical conductivity, the modified rare-earth perovskite-based catalysts deliver better electrocatalytic performance than the pristine perovskite and are even superior to that of noble metals-based compounds. However, although considerable development has been made on rare-earth perovskite-based catalysts in recent years, suggestions for improving the performance further and a deeper understanding of the electrocatalytic mechanism of perovskite electrocatalysts are also essential and described in the following.

(i) The traditional preparation method for perovskites requires high-temperature sintering, which is high energy consumption and leads to large particle size with a small specific surface area for electrocatalysis. To extend the application of the rare-earth-based perovskites, green and low-temperature synthesis techniques should be developed. For example, sol-gel, hydrothermal, solution method, etc. should be developed to be in line with the requirement of the low-carbon economy. In addition, perovskite electrocatalysts with porous structures and a large specific surface area to have more active site exposure are required for high-performance OER in the future.
(ii) The previous works reported that A-site substitution affects the valence state of B-site ions so that to modulate the electronic configuration and the crystal structure, whereas the B-site substitution tunes the electronic configuration and charge transport efficiency of the electrocatalyst. Therefore, it is proposed that the co-substitution of the A/B-site can modulate the electronic structure and charge transport performance precisely, as a result, higher performed OER catalysts would be developed.
(iii) Composite engineering is effective to construct multifunctional electrocatalysts, where the interface between the two components should have played a significant role in the superior catalytic performance. Therefore, the interface should be well identified and understand the catalytic mechanism deeply. Additionally, the revolution of the interface during the OER should be monitored using *in situ* methods including *in situ* XRD, *in situ* TEM, *in situ* XAS (synchrotron X-ray absorption spectroscopy), etc., which are important to reveal the catalyst evolution during the reactions. For instance, *in situ* XRD can be used to detect the crystal structure change (crystal or amorphous, crystal form, lattice parameters, etc.) of the rare-earth-based perovskites catalysts during the reaction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was financially supported by the National Natural Science Foundation of China (52002294), Key Research and Development Program of Hubei Province (2021BAA208),

Outstanding Youth Foundation of Natural Science Foundation of Hubei Province (2020CFA099), and Knowledge Innovation Program of Wuhan-Shuguang Project (2022010801020364).

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